



Lab Task 02, 03

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Subject	AI

1. Sum of List Elements



```
def sum_list(lst):  
    total = 0  
    for num in lst:  
        total += num  
    return total
```

```
numbers = [1, 2, 3, 4, 5]  
print(sum_list(numbers)) |
```

Output

```
The sum is:  
15
```

2. Count Even and Odd Numbers

 

```
def count_even_odd(lst):  
    even_count = 0  
    odd_count = 0  
  
    for num in lst:  
        if num % 2 == 0:  
            even_count += 1  
        else:  
            odd_count += 1  
  
    print("Even numbers:", even_count)  
    print("Odd numbers:", odd_count)  
  
numbers = [1, 2, 3, 4, 5, 6, 7]  
count_even_odd(numbers)
```

Output



```
Even numbers: 3  
Odd numbers: 4
```

3. Find Maximum in a List

```
 def find_max(lst):  
    if not lst:  
        return None  
  
    max_num = lst[0]  
    for num in lst:  
        if num > max_num:  
            max_num = num  
    return max_num  
  
numbers = [3, 7, 2, 9, 1, 4]  
print(find_max(numbers))
```

Output

```
 9
```

4. Reverse a List

```
 def reverse_list(lst):  
    reversed_lst = []  
    for i in range(len(lst) - 1, -1, -1):  
        reversed_lst.append(lst[i])  
    return reversed_lst  
  
numbers = [1, 2, 3, 4, 5]  
reversed_numbers = reverse_list(numbers)  
  
print("Original list:", numbers)  
print("Reversed list:", reversed_numbers)
```

Output

```
 Original list: [1, 2, 3, 4, 5]  
Reversed list: [5, 4, 3, 2, 1]
```

5. Check Prime Number

```
 def is_prime(n):  
    if n <= 1:  
        return False  
    for i in range(2, int(n ** 0.5) + 1):  
        if n % i == 0:  
            return False  
    return True  
  
print(is_prime(2))  
print(is_prime(7))  
print(is_prime(10))  
print(is_prime(1))  
print(is_prime(23))
```

Output

```
 True  
True  
False  
False  
True
```

6. Factorial Using While Loop



```
def factorial(n):  
    result = 1  
    while n > 1:  
        result *= n # Multiply result by current n  
        n -= 1      # Decrease n by 1  
    return result  
  
print(factorial(5)) # Output: 120  
print(factorial(0)) # Output: 1
```

Output



```
120  
1
```

7. Fibonacci Sequence

```
 def fibonacci(n):  
  
    if n <= 0:  
        print("Please enter a positive integer")  
        return  
  
    a, b = 0, 1  
  
    print(f"First {n} Fibonacci numbers:")  
    for _ in range(n):  
        print(a, end=' ')  
        a, b = b, a + b  
    print()  
  
fibonacci(10)
```

Output

```
 First 10 Fibonacci numbers:  
0 1 1 2 3 5 8 13 21 34
```


8. Remove Duplicates from a List

```
 def remove_duplicates(lst):  
    unique_list=[]  
    for item in lst:  
        if item not in unique_list:  
            unique_list.append(item)  
    return unique_list  
  
numbers=[1,1,2,3,4,4,5,6,6,7,7,8,9]  
print("Duplicated List:", numbers)  
print("Deduplicated List:", remove_duplicates(numbers))
```

Output

```
 Duplicated List: [1, 1, 2, 3, 4, 4, 5, 6, 6, 7, 7, 8, 9]  
Deduplicated List: [1, 2, 3, 4, 5, 6, 7, 8, 9]
```

9. Find Common Elements in Two Lists

```
 def common_elements(lst1, lst2):  
    common = []  
    for item in lst1:  
        if item in lst2 and item not in common:  
            common.append(item)  
    return common  
  
list1 = [1, 2, 3, 4, 5]  
list2 = [3, 4, 5, 6, 7]  
print("Common elements:", common_elements(list1, list2))
```

Output

 Common elements: [3, 4, 5]

10. Palindrome Check

```
 def is_palindrome(s):  
    s = s.lower().replace(" ", "")  
    return s == s[::-1]  
  
print(is_palindrome("madam"))           # True  
print(is_palindrome("racecar"))         # True  
print(is_palindrome("hello"))           # False  
print(is_palindrome("A man a plan a canal Panama")) # True
```

Output

 True
True
False
True